

Users and Passwords:

User: clerk (Lowest level employee only able to see data) → Check data Ex: See class rosters (This could be the “teacher role”)

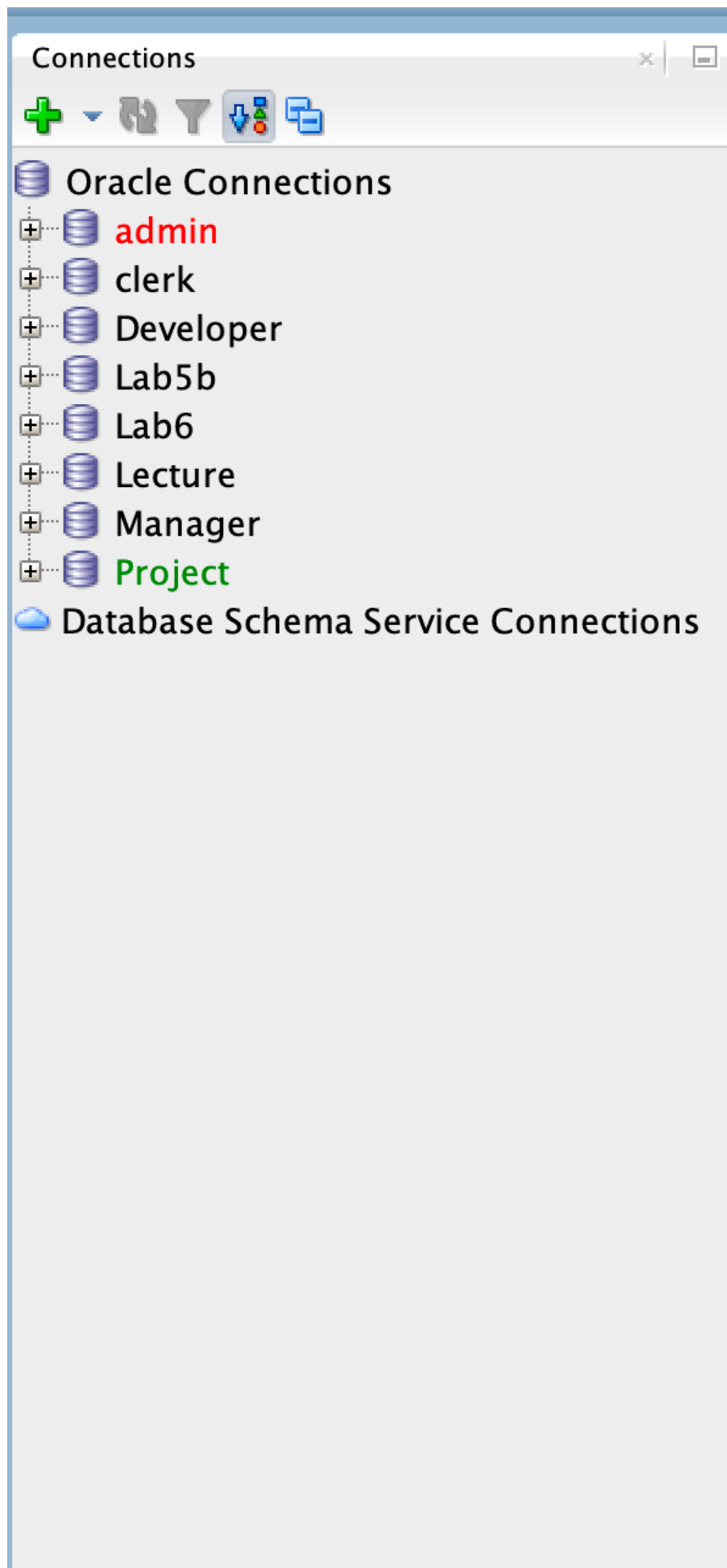
Password: 02GsaG0FtmTCKbiiRJhT

User: manager (Mid-level employee able to see, add, update, and delete data) → Check, Update, delete data Ex: Enroll Students in classes (This could be the advisor role)

Password: n67na3hoo85jGJhqDkYV

User: developer (High-level employee able to see, add , update and delete data and create new database objects) Check, Update, delete data and add tables, views, etc. Ex: My role in this project creating tables and adding data is the role that I play in this company

Password: 7qjBiL23mD3qx0WGMhyF



Clerk Screenshots:

The screenshot shows the Clerk application interface with the 'Query Builder' tab active. The SQL editor contains the following code:

```
-- Clerk Privileges  
  
-- Pass  
Select *  
from project.student;  
  
-- FAIL  
Insert Into project.Building VALUES(2,'Test Building');
```

The 'Query Result' tab is selected, displaying the results of the first query. The status bar indicates 'All Rows Fetched: 5 in 0.04 seconds'.

	STUDENT_ID	FIRST_NAME	LAST_NAME	DATE_OF_BIRTH	GRADE_LEVEL	EMAIL	PHONE	ADVISOR_FACULTY_ID
1	1	korey	rodi	13-MAR-06		12 rodik@gmail.com	856780210	5
2	2	Cassidy	Coomey	19-JAN-06		11 cooney@yahoo.com	8672220987	6
3	3	Colin	Burns	02-DEC-05		10 burns@aol.com	3671234567	10
4	4	Mitchell	Dalesio	17-JAN-06		9 DalesioM@gmail.com	2678902130	10
5	5	Sean	Gallagher	03-NOV-07		9 GallagherS@gmail.com	8567894560	5

The screenshot shows the Clerk application interface with the 'Query Builder' tab active. The SQL editor contains the same code as the first screenshot:

```
-- Clerk Privileges  
  
-- Pass  
Select *  
from project.student;  
  
-- FAIL  
Insert Into project.Building VALUES(2,'Test Building');
```

The 'Script Output' tab is selected, displaying an error message. The status bar indicates 'Task completed in 0.107 seconds'.

```
Error starting at line : 10 in command -  
Insert Into project.Building VALUES(2,'Test Building')  
Error at Command Line : 10 Column : 21  
Error report -  
SQL Error: ORA-01031: insufficient privileges  
  
https://docs.oracle.com/error-help/db/ora-01031/01031. 00000 - "insufficient privileges"  
*Document: YES  
*Cause: A database operation was attempted without the required  
privilege(s).  
*Action: Ask your database administrator or security administrator to grant  
you the required privilege(s).  
  
More Details :  
https://docs.oracle.com/error-help/db/ora-01031/
```

Manager Screenshots:

The screenshot displays the Oracle SQL Developer Manager interface. The top pane, titled 'Run Statement (^+↵) | der', shows a SQL script with the following content:

```
-- Manager Privileges  
  
-- Pass  
Select * from project.student;  
  
INSERT INTO project.building VALUES(2,'test');  
  
-- FAIL  
Create table test (  
    testcol integer not null  
);
```

The bottom pane, titled 'Query R...', shows the execution results for the first statement. It indicates 'All Rows Fetched: 5 in 0.037 seconds'. The results are displayed in a table with the following columns: STUDENT_ID, FIRST_NAME, LAST_NAME, DATE_OF_BIRTH, GRADE_LEVEL, EMAIL, PHONE, and ADVISOR_FACULTY_ID.

	STUDENT_ID	FIRST_NAME	LAST_NAME	DATE_OF_BIRTH	GRADE_LEVEL	EMAIL	PHONE	ADVISOR_FACULTY_ID
1	1	korey	rodi	13-MAR-06		12 rodik@gmail.com	856780210	5
2	2	Cassidy	Coomey	19-JAN-06		11 cooney@yahoo.com	8672220987	6
3	3	Colin	Burns	02-DEC-05		10 burns@aol.com	3671234567	10
4	4	Mitchell	Dalesio	17-JAN-06		9 DalesioM@gmail.com	2678902130	10
5	5	Sean	Gallagher	03-NOV-07		9 GallagherS@gmail.com	8567894560	5

Manager Developer | 0.046 seconds

Worksheet Query Builder

Manager Privileges

Pass

```
Select * from project.student;
```

INSERT INTO project.building VALUES(2,'test');

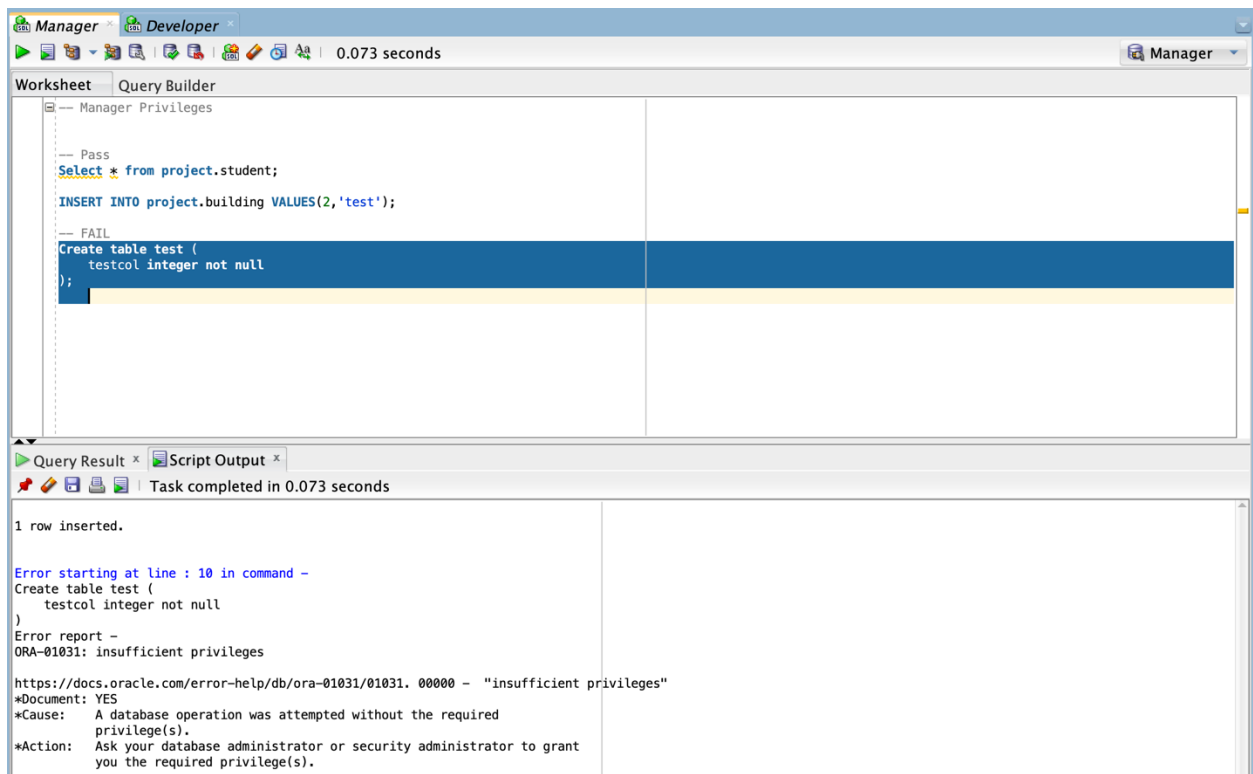
FAIL

```
Create table test (  
  testcol integer not null  
);
```

Query Result x Script Output x

Task completed in 0.046 seconds

1 row inserted.



Developer Screenshots:

Developer

Worksheet | Query Builder

Developer Privileges

Pass

```
Select * from project.student;  
  
INSERT INTO project.building VALUES(3,'test');  
  
Create table test (  
  testcol integer not null  
);
```

Query Result x

SQL | All Rows Fetched: 5 in 0.095 seconds

	STUDENT_ID	FIRST_NAME	LAST_NAME	DATE_OF_BIRTH	GRADE_LEVEL	EMAIL	PHONE	ADVISOR_FACULTY_ID
1	1	korey	rodi	13-MAR-06		12.rodik@gmail.com	856780210	5
2	2	Cassidy	Coomey	19-JAN-06		11.cooney@yahoo.com	8672220987	6
3	3	Colin	Burns	02-DEC-05		10.burnsC@aol.com	3671234567	10
4	4	Mitchell	Dalesio	17-JAN-06		9.DalesioM@gmail.com	2678902130	10
5	5	Sean	Gallagher	03-NOV-07		9.GallagherS@gmail.com	8567894560	5

Developer

Worksheet | Query Builder

Developer Privileges

Pass

```
Select * from project.student;  
  
INSERT INTO project.building VALUES(3,'test');  
  
Create table test (  
  testcol integer not null  
);
```

Query Result x | Script Output x

Task completed in 0.034 seconds

1 row inserted.

